



Multiphysics model of thermomechanical oxidative degradation in SiC/SiC ceramic matrix composite microstructures

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ABSTRACT

A coupled thermomechanical-oxidation damage formulation, based on governing multiphysics constitutive relations and stochastic microstructure representation, is developed to provide fundamental insights into the life-limiting oxidative degradation mechanisms in ceramic matrix composites (CMCs) with a silicon carbide (SiC)-based matrix reinforced by SiC fibers. The methodology includes passive oxidation of the fibers governed by reaction kinetics and diffusion-limited degradation. Conservation of mass equations, incorporating a modified Fick's law, govern the evolution of oxygen and silica concentrations in CMC constituents. The effective oxygen diffusivities in the matrix and fibers are modeled as functions of matrix damage and normalized silica concentration, respectively. The model is numerically implemented using the finite element method and tested on stochastic representative volume elements, accounting for the inherent flaws in the CMC microstructure. The fundamental differences between oxidation regimes, the time-dependent deformation mechanisms, and the influence of matrix damage on oxidation kinetics at the microscale are analyzed.

1. Introduction

Ceramic matrix composites (CMCs) offer higher temperature and lower density alternatives to metallic superalloys for a range of applications, including aerospace, nuclear, deep space, and defense [1–4]. High temperature CMCs used in primary structures often comprise silicon carbide (SiC)-based matrix reinforced by SiC fibers and a weak fiber/matrix interphase, such as boron nitride (BN) interphase, designed to impart damage tolerance and pseudo ductility to an otherwise brittle material. In these extreme operating environments, CMC components experience high temperatures, thermal gradients, sustained loading, and transients. For example, during combustion, the oxidative environments involve prolonged exposure to elevated temperatures and water vapor that result in time and temperature-dependent embrittlement and subsequent loss of composite strength. Manufacturing processes such as polymer infiltration and pyrolysis (PIP) introduce multiscale defects in the matrix and expose SiC fibers to such environments where they degrade at different rates depending on the degree of fiber stoichiometry [5,6]. Therefore, to retain structural integrity for varying time periods and environmental conditions, it is essential to prevent SiC fiber oxidation and silica volatilization which lead to excessive surface recession and thermomechanical performance degradation. Hence, it is

crucial to understand the rate-dependent thermomechanical-oxidative stability that governs the life of CMC components in such extreme environments.

The deleterious effect of oxidizing environments on high-performance SiC fiber-based CMC's mechanical behavior at elevated temperatures has been well documented [7–11]. Oxidation-assisted creep and stress rupture were observed at intermediate and high temperatures [6,12–14]. At intermediate temperatures, under thermomechanical load with sufficient stress for opening the matrix microcracks, woven SiC/SiC CMC exhibits severe embrittlement and premature fracture, which is a phenomenon known as “peeling” [8], as well as time-dependent inelastic deformation [15,16]. Stress rupture experiments on PIP CMCs showed that the main oxidation mechanisms are internal and surface oxidation of fibers, and fiber/matrix interphase oxidation [17]. Progressive matrix microcracking exposes fibers to the oxidizing environment, thereby accelerating intrinsic fiber degradation [18–21]. Oxygen diffuses through the matrix microcracks and reacts with BN interphase coating and SiC fibers, forming gaseous species and brittle borosilicate reaction products. The solid borosilicate reaction products can act as a bonding agent, effectively bonding exposed fibers together at the sites of matrix cracks. Since the fused fibers cannot move or slip independently, localized stress concentration occurs at the fiber

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surfaces, leading to brittle and catastrophic CMC failure. Further, the phase transformation of SiC fiber during oxidation results in nonuniform volumetric changes and induced tensile stresses, leading to fiber strength reduction [22]. However, at high temperatures (≥ 1200 °C) under unstressed oxidation conditions, the SiC matrix oxidizes to form molten oxide ceramics, SiO₂, also known as silica. Consequently, this seals the oxygen diffusive channels and protects the fiber and the BN interphase from oxidation and subsequent embrittlement [10]. Additionally, the coupling between damage and oxidation at high temperatures under stressed oxidation tests causes the CMCs to exhibit creep-like strain behavior, eventually leading to tertiary creep and subsequent failure [23,24]. Finally, at ultra-high temperatures (≥ 1600 °C), the deformation mechanisms and the CMC life are dominated by fiber creep [13,16,25,26]. Reaction-limited oxidation can occur at intermediate temperatures, where the oxygen diffusivity is significantly faster than the chemical reaction rate. Oxidation occurs simultaneously at all the fiber filaments within the tow and results in significant change in the oxygen concentration. Conversely, at high temperatures, diffusion-limited oxidation was mainly observed, where the temperature is high enough to rapidly trigger the fiber and interphase chemical reactions, resulting in the oxidation being limited to the diffusive channels within the matrix [23,27]. However, in practice, the oxidation behavior is a combination of both regimes [18]. Even though significant efforts have been done to investigate the oxidation of monolithic SiC and CMCs under lab air, dry oxygen and steam oxidation conditions [18,28–32], there is a gap in the literature regarding the effect of matrix microcracks and damage on the oxidation mechanisms of stressed SiC/SiC materials.

Various theoretical formulations have been proposed to model the oxidation of CMCs in service environments. Orthotropic continuum damage model was developed to model oxidation-assisted Hi-Nicalon fiber strength degradation based on slow crack growth [6]. The model assumed that the degree of fiber exposure to oxygen depends on the matrix damage, while neglecting the complex oxygen kinetics within the matrix diffusive channels and the subsequent chemical reactions of the microstructure constituents. Analytical one-dimensional model was developed to mimic the silica formation due to SiC fiber oxidation at intermediate temperatures, as well as the resulting induced tensile stresses in the fiber [33]. The model has not been extended to multiaxial stress state and lacks the oxidative sealing of fibers and matrix cracks. Micromechanics models [13] for the primary rate-determining phenomena of SiC/SiC CMC oxidation were developed, including the diffusion of oxidizing species and reaction products, fiber oxidation and the loss of stiffness, partial matrix microcrack sealing and boron nitride coating loss at the fiber surfaces. These models are only valid for simple idealized geometries and neglect the time-dependent thermomechanical response of fibers due to creep. Zhao et al. [34] modeled oxidation-damage coupling with oxygen diffusion through micro- and mesoscale architectures using finite element analysis (FEA) simulations. However, the formulation is restricted to the diffusion-controlled regime, lacks key considerations for heterogeneous diffusion, and the studies were confined to idealized geometries. Yu et al. [35] developed an equivalent diffusion coefficient model to simulate oxidation kinetics in SiC/pyrolytic carbon (PyC)/SiC CMCs, establishing a relationship between effective oxygen diffusivity and material porosity, matrix crack density, and matrix crack width. However, the model employed a first-order chemical reaction rate, steady-state mass transfer for oxygen diffusivity, and a homogenization approach for the manufacturing-induced defects, which limits its implementation into multiphysics multiscale modeling frameworks. A multiphysics model was developed for SiC fiber oxidation, accounting for diffusion, oxidation reaction rates, and viscoelastic deformation [22]. However, this model is limited to a single SiC fiber and does not account for the two-way coupling between oxidation and damage. Finally, Schichtel and Chattopadhyay [23] developed a multiphysics damage and oxidation model for heterogeneous C/SiC CMC microstructures. This work was focused on high temperature oxidative regime; therefore, the SiC

oxidation and the subsequent silica formation are not considered.

In this work, a novel multiphysics model is developed to simulate the multi-regime oxidation of PIP CMCs. The goal is to investigate the fundamental differences between diffusion-limited and reaction-limited oxidation, as well as oxidation-induced inelastic strain. The governing equations for equilibrium and conservation of mass are linked through chemical reaction terms, matrix damage, and oxygen diffusivity and solubility to address the spatiotemporal response of CMC microstructures at elevated temperatures. The model is discretized and implemented into FEA to simulate both unstressed and stressed (creep) oxidation tests. A recently developed deep learning (DL) algorithm for stochastic representative volume element (SRVE) generation [36–38] is used in the numerical simulations to accurately investigate the effect of microstructure variability and intratow porosity on the oxidation kinetics. The analysis aims to address the following question: how do different time-dependent mechanisms, such as gaseous diffusion, rate-limited reactions, and damage, interact and impact the spatiotemporal response of CMC microstructures?

2. Multiphysics model formulation

The derivation of the multiphysics model that can capture the complex coupled oxygen diffusion, CMC constituents' chemical reactions, damage, and the resulting creep-like strain is based on the governing equations for force equilibrium and conservation of mass for oxygen and silica. In the formulations, boldface symbols and letters denote vectors and matrices. The superscripts denote that the quantity of interest corresponds to a certain microstructure constituent (i.e., matrix, fiber, and porosity).

2.1. Constitutive model

Quasistatic loading is assumed, which results in neglecting the inertia effects, and the equilibrium equation, in terms of the Cauchy stress tensor (σ), and in the absence of body forces is given as:

$$\nabla \cdot \sigma = 0 \quad (1)$$

Constitutive models are applied to the CMC microstructure, including matrix, fiber, and porosity, accounting for matrix damage due to microcrack evolution and fiber stiffness degradation resulting from silica formation at the SiC fiber surfaces. This is expressed as follows.

$$\sigma^M = (1 - d)C_0^M \epsilon \quad (2a)$$

$$\sigma^F = [\phi_{SiO_2} C_0^{SiO_2} + (1 - \phi_{SiO_2}) C_0^F] \epsilon \quad (2b)$$

$$\sigma^{porosity} = C_0^{porosity} \epsilon \quad (2c)$$

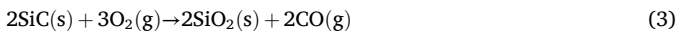
where C_0 is the initial fourth order stiffness tensor corresponding to each constituent, d is a scalar matrix damage variable, ϵ is the total mechanical strain tensor, and ϕ_{SiO_2} is the normalized silica concentration. Infinitesimal strain theory is assumed due to the high stiffness of the primary constituents. For each simulation the temperature is assumed to be constant throughout, and the exothermic heating of the reaction is not included, i.e., effects of temperature gradients are not considered. The damage parameter, d , can be governed by any material damage model, such as fracture mechanics-based models [39–41]. However, in this work, a simple exponential softening law based on the damage energy release rate, which is a function of strain is used [34,42]. This simplification allows for accurate computation of the consistent tangent modulus necessary for the FEA numerical implementation as discussed Section 3.1. The initial stiffness of porosity is selected to be a small arbitrary non-singular tensor to reflect the pre-damaged state of the SiC matrix because of manufacturing-induced defects. To compute the SiC fiber stiffness evolution with silica formation, a linear interpolation approach, which is generally used in multifield coupled models [22], is

adopted. The stiffness linearly depends on the normalized concentration of silica which is attributed to SiC fiber oxidation degree as shown in Eq. (2b). The normalized concentration is defined as $\phi = \frac{c}{s}$, where c is the molar concentration and s is the solubility of the oxidizing species and the chemical reaction products in each medium. The solubility of the material is defined as the capacity or the equilibrium concentration of a given element in the material for a given state. The normalized silica concentration range ($0 \leq \phi_{\text{SiO}_2} \leq 1$) postulates that when the SiC fiber is pristine, $\phi_{\text{SiO}_2} = 0$, and $\phi_{\text{SiO}_2} = 1$ indicates that the maximum concentration of silica in SiC fiber has been reached, and no further chemical reaction can take place.

This work addresses the oxidation kinetics of a non-oxide PIP SiC/SiC CMC with a matrix consisting of Si, N, and C (referred as silicon nitro carbide, SiNC). Detwiler et al. [18] studied the oxidation of chemical vapor infiltration (CVI) SiC/BN/SiC CMCs using thermogravimetric analysis (TGA) in various oxidizing environments and temperatures. The B-rich phase in the hybrid CVI and melt infiltrated (MI) matrix, as well as the CMC surface, oxidizes to form a protective silica layer, sealing the fiber. However, in PIP SiC/SiC CMCs, the unreacted preceramic polymer with excess C and Si decomposes at elevated temperatures [43,44]. In-house TGA testing of PIP CMC shows that these decomposition reactions result in mass loss at 1200 °C, unlike its CVI counterpart. Our initial analysis showed that mass gain due to silica formation and mass loss due to thermally induced decomposition reactions in SiNC matrix are two competing mechanisms. Hence, the assumption of neglecting the silica formation in the matrix for this study will not significantly affect the SiC fiber oxidation kinetics. Regardless, the model will provide accurate estimation of the critical silica thickness and mass at the SiC fiber surface in the absence of silica damage. The hypotheses of the model are: i) only SiC fiber oxidation—which has been reported in the literature [45] as the driving mechanism—is considered; and ii) fiber degradation is attributed to modulus loss. The phase transformation of SiC to silica is accompanied by a large volumetric change (~220 %), where nonuniform expansion around SiC fibers generates significant axial and hoop stresses on the order of GPa [19,46]. These stresses induce microcracks in the silica thin film, affecting both SiC fiber strength and oxidation kinetics. Furthermore, high dislocation densities in crystalline silica have been observed, supporting the existence of significant shear stresses at the SiC-silica interface [45]. Zhao et al. [22] examined the influence of volumetric changes on oxygen diffusivity, but fiber strength degradation and silica fracture were not incorporated into their analysis. Based on the above discussion, SiC fiber oxidation in PIP CMCs is the driving mechanism in thermomechanical performance degradation. Therefore, the consideration of only SiC fiber oxidation in PIP CMCs for both unstressed and stressed oxidation tests is reasonable. When integrated into structural-scale simulations, the developed model will enhance the fidelity of thermomechanical life prediction for CMC components operating in harsh environments [40,47,48].

2.2. Oxidation kinetics

At intermediate and high temperatures, SiC can undergo either active or passive oxidation. At low oxygen partial pressure (<1 atm), active oxidation occurs which results in SiC volatilization. However, passive oxidation is typically observed in SiC/SiC CMCs under dry oxygen and engine operating environments [17,20], and is therefore considered in this work. In passive oxidation, the SiC fiber oxidizes and forms a protective silica film on the fiber surface according to:



To track the oxidizing species and the chemical reaction products, the conservation of mass equations are employed as follows.

$$\frac{dc_{\text{O}_2}}{dt} + \nabla \cdot \mathbf{j}_{\text{O}_2} = -3\xi \quad (4a)$$

$$\frac{dc_{\text{SiO}_2}}{dt} = 2\xi \quad (4b)$$

where $\mathbf{j}$ is the molar flux, ξ is the chemical reaction rate, and t is the time. The effects of CO are ignored in this study, since the pressure gradient-driven Darcy flow was determined to have minor impact on the CMC oxidation [49]. Additionally, the flux term is only included in the oxygen conservation of mass equation, since the silica is in a viscous form and rapidly solidifies at elevated temperatures.

The diffusion flux is defined using the Fick's law as follows.

$$\mathbf{j}_{\text{O}_2} = -\frac{D_{\text{O}_2} c_{\text{O}_2}}{RT} \frac{\partial \mu_{\text{O}_2}}{\partial \mathbf{x}} \quad (5a)$$

$$\mu_{\text{O}_2} = \mu_{\text{O}_2}^0 + RT \ln \phi_{\text{O}_2} \quad (5b)$$

where D_{O_2} is the effective oxygen diffusivity, $\mu_{\text{O}_2}^0$ and μ_{O_2} are the reference and total chemical potentials, respectively, R is the gas constant, and T is the temperature. Generally, the diffusivity is a second order tensor; however, due to the lack of available data, it is assumed to be isotropic, with D_{O_2} represented as a scalar value. Regardless, the overall diffusivity of the CMC microstructure is highly anisotropic due to the implementation of high-fidelity SRVE, as discussed in Section 3.2.

According to point defect kinetics [50], the driving force of diffusion is a function of the total chemical potential, which is approximated by Eq. (5b) based on a chemo-thermo-mechanically coupled theory for elastic and viscoplastic deformation [51]. This equation results in approximating the analytical flux equation for heterogeneous medium that depends on the gradient of current and equilibrium concentrations. Note that in case of homogeneous diffusive media, the gradient of the chemical potential can be expressed as the gradient of the current oxygen concentration ($\frac{\partial c_{\text{O}_2}}{\partial \mathbf{x}}$) [34]. However, since CMC microstructures are highly heterogeneous, Eq. (5a) is implemented for accurate spatiotemporal solution of oxygen diffusion. Using Eq. (5), the modified Fick's law for oxygen diffusivity can be formulated as follows.

$$\mathbf{j}_{\text{O}_2} = -D_{\text{O}_2} s_{\text{O}_2} \frac{\partial \phi_{\text{O}_2}}{\partial \mathbf{x}} \quad (6)$$

Note that Eq. (6) agrees with Schichtel and Chattopadhyay [23], where the flux equation includes an additional solubility term that reconciles the flux units while accounting for the heterogeneity of the microstructure.

The chemical energy (ψ^{chem}) that represents the mixture of SiC and SiO₂ is assumed to follow a quadric form as discussed by Drozdov et al. [52].

$$\psi^{\text{chem}}(c_{\text{SiO}_2}) = \frac{1}{2} k (c_{\text{SiO}_2}^{\text{max}} - c_{\text{SiO}_2})^2 \quad (7)$$

where k is a positive coefficient, analogous to elastic modulus in strain energy density, and $c_{\text{SiO}_2}^{\text{max}}$ is the maximum concentration of silica in SiC reference configuration (i.e., molar concentration in reference configuration upon chemical reaction completion). As shown by Zhao et al. [22], to have a non-negative dissipation which satisfies the second law of thermodynamics, the chemical reaction rate should be governed by:

$$\xi = k c_{\text{O}_2} (c_{\text{SiO}_2}^{\text{max}} - c_{\text{SiO}_2}) K_f \quad (8a)$$

$$K_f = K_0 \exp\left(\frac{-Q_\xi}{RT}\right) \quad (8b)$$

where K_f is the coefficient of the chemical reaction rate, Q_ξ and K_0 are the activation energy of the chemical reaction and the reaction rate constant, respectively. K_f is defined based on the Boltzmann's distribution concept, where it has an Arrhenius temperature dependency. Eq. 8 implies that as the chemical reaction proceeds and silica evolves, the chemical reaction rate drops, thereby mimicking the sealing of SiC fiber

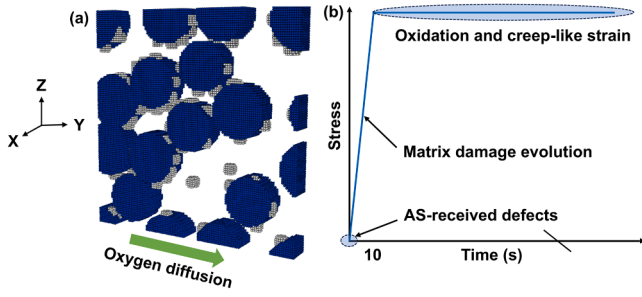


Fig. 1. Numerical simulation setup; (a) A sample of three-dimensional stochastic representative volume element (SRVE) for SiC/SiC CMC with uniform structural mesh, showing SiC fiber elements in blue and porosity elements in off-white; (b) Applied loading for stressed oxidation simulations in the axial fiber direction (X-direction) at elevated temperatures.

Table 1
SiC matrix, Hi-Nicalon™ fiber and SiO₂ elastic properties [22,38–41].

	E (GPa)	ν	σ_{UTS} (MPa)
SiC matrix	420	0.14	180
Hi-Nicalon™ fiber	200	0.22	-
Silica (SiO ₂)	70	0.17	-

Table 2
Multiphysics oxidation model parameters [5,22,23].

Symbol	Value
R	8.314 J mol ⁻¹ K ⁻¹
$c_{SiO_2}^{max}$	8.04×10^4 mol m ⁻³
$c_{O_2}^{max}$	0.19 mol m ⁻³
Q_2	25 KJ mol ⁻¹
kK_0	2×10^{-1} m ³ mol ⁻¹ s ⁻¹
Q_{O_2}	82 KJ mol ⁻¹
$D_{O_2}^0$	6×10^{-10} m ² s ⁻¹
χ_{O_2}	0.20946
α_1	1×10^{-3}
α_2	1×10^{-3} (case 1); 1×10^{-2} (case 2)
M_1	31.99 g mol ⁻¹
M_2	28.96 g mol ⁻¹
M_{SiC}	40.11 g mol ⁻¹
M_{SiO_2}	60.08 g mol ⁻¹
σ_1	3.46 Å
σ_2	3.62 Å

and the experimentally observed para-linear oxidation kinetics [5,53].

The local oxygen capacity and diffusivity are directly coupled to the state of damage in the matrix and the state of oxidation in the fibers. Following the basic principles of porous media theory, oxygen is assumed to be present only in the pores and cracks of the matrix and not within the physical solid matrix material. Hence, the matrix oxygen solubility evolves with the damage parameter. Additionally, the oxygen diffusivity in silica is significantly higher than that in SiC fiber [22]. Therefore, the oxygen diffusivity in fibers depends on the concentration of silica formed along their periphery. In this work, the oxygen solubility in matrix and fiber is assumed to vary linearly as follows.

$$s_{O_2}^M = (s_{maxO_2}^M - s_{minO_2}^M)d + s_{minO_2}^M \quad (9a)$$

$$s_{O_2}^F = (s_{maxO_2}^F - s_{minO_2}^F)\phi_{SiO_2} + s_{minO_2}^F \quad (9b)$$

where s_{minO_2} and s_{maxO_2} are the minimum and maximum capacities of oxygen in the material, respectively. The maximum solubility in matrix ($s_{maxO_2}^M$) is the concentration of oxygen in air at the simulated temperature (s_{maxO_2}) which corresponds to a totally damaged matrix, while the

minimum solubility in matrix ($s_{minO_2}^M$) is the capacity of oxygen in the pre-existing diffusive channels due to manufacturing-induced defects, this is selected to be a small percentage of s_{maxO_2} , as shown in Eq. (10b), to allow an initial oxygen ingress into the microstructure. The effect of initial defected matrix on oxidation mechanisms is discussed in Section 4.1. Meanwhile, by definition, the maximum oxygen solubility in fiber ($s_{maxO_2}^F$) corresponds to the maximum oxygen concentration in the silica scale, denoted as $c_{O_2}^{max}$, and the minimum oxygen solubility in fiber ($s_{minO_2}^F$) is selected to be a small arbitrary number to trigger the chemical reaction. Finally, the oxygen solubility in porosity ($s_{O_2}^{porosity}$) is set to the oxygen concentration in air, similar to $s_{maxO_2}^M$, where it is assumed to be constant in the absence of thermal gradients.

$$s_{minO_2}^F = \alpha_1 c_{O_2}^{max} \quad (10a)$$

$$s_{minO_2}^M = \alpha_2 s_{maxO_2} \quad (10b)$$

$$s_{maxO_2} = \chi_{O_2} \frac{P}{RT} \quad (10c)$$

where χ_{O_2} is the molar fraction of oxygen in air and P is the ambient pressure. α_1 and α_2 are the fractions for the minimum oxygen solubility/diffusivity in fiber and matrix respectively.

The local oxygen diffusivities (D_{O_2}) are comparably defined as:

$$D_{O_2}^M = (D_{maxO_2}^M - D_{minO_2}^M)d + D_{minO_2}^M \quad (11a)$$

$$D_{O_2}^F = (\phi_{SiO_2})^2 D_{O_2}^0 \exp\left(-\frac{Q_{O_2}}{RT}\right) + D_{minO_2}^F \quad (11b)$$

$$D_{minO_2}^M = \alpha_2 D_{maxO_2} \quad (11c)$$

$$D_{maxO_2} = \frac{0.001858}{P\sigma_{12}^2\Omega_D} \sqrt{T^3 \frac{(M_1 + M_2)}{M_1 M_2}} \quad (11d)$$

where $D_{minO_2}^M$ is the minimum oxygen diffusivity in matrix, which is defined as a fraction of the maximum oxygen diffusivity $D_{maxO_2}^M$. $D_{maxO_2}^M$ is set to the oxygen diffusivity in air D_{maxO_2} . Following the kinetic theory of gases [54], D_{maxO_2} depends on ambient pressure, temperature, collision integral Ω_D , the collision diameter σ_{12} , and the molar masses of species 1 (oxygen) and species 2 (air), denoted as M_1 and M_2 , respectively. σ_{12} can be calculated from the effective diameter of each species $\sigma_{12} = \frac{1}{2}(\sigma_1 + \sigma_2)$. Ω_D depends on temperature, hence its values are interpolated from the corresponding tabulated values [54]. Due to the inherent oxygen diffusivity in silica, Eq. (11b) links the fiber oxygen diffusivity to the normalized silica concentration on the fiber surfaces, as well as to the diffusion constant $D_{O_2}^0$, the diffusion activation energy Q_{O_2} , and temperature. Like the minimum oxygen solubility in fiber, $D_{minO_2}^F$ is selected to be an arbitrary small number to trigger oxygen diffusion and the subsequent chemical reaction. Finally, the oxygen diffusivity in porosity is set to D_{maxO_2} .

3. Numerical implementation

3.1. Finite element modeling

The multiphysics model is implemented numerically using FEA via Abaqus. The model has five degrees of freedom: the three displacement components $\mathbf{u}$, the normalized oxygen concentration ϕ_{O_2} , and the normalized reaction product concentration ϕ_{SiO_2} . The implementation of this model into FEA software requires the usage of a user-defined element (UEL) subroutine, which is a customized finite element subroutine for solving the problem with multiphysics degrees of freedom. Structured orthogonal mesh is used with trilinear hexahedral eight node

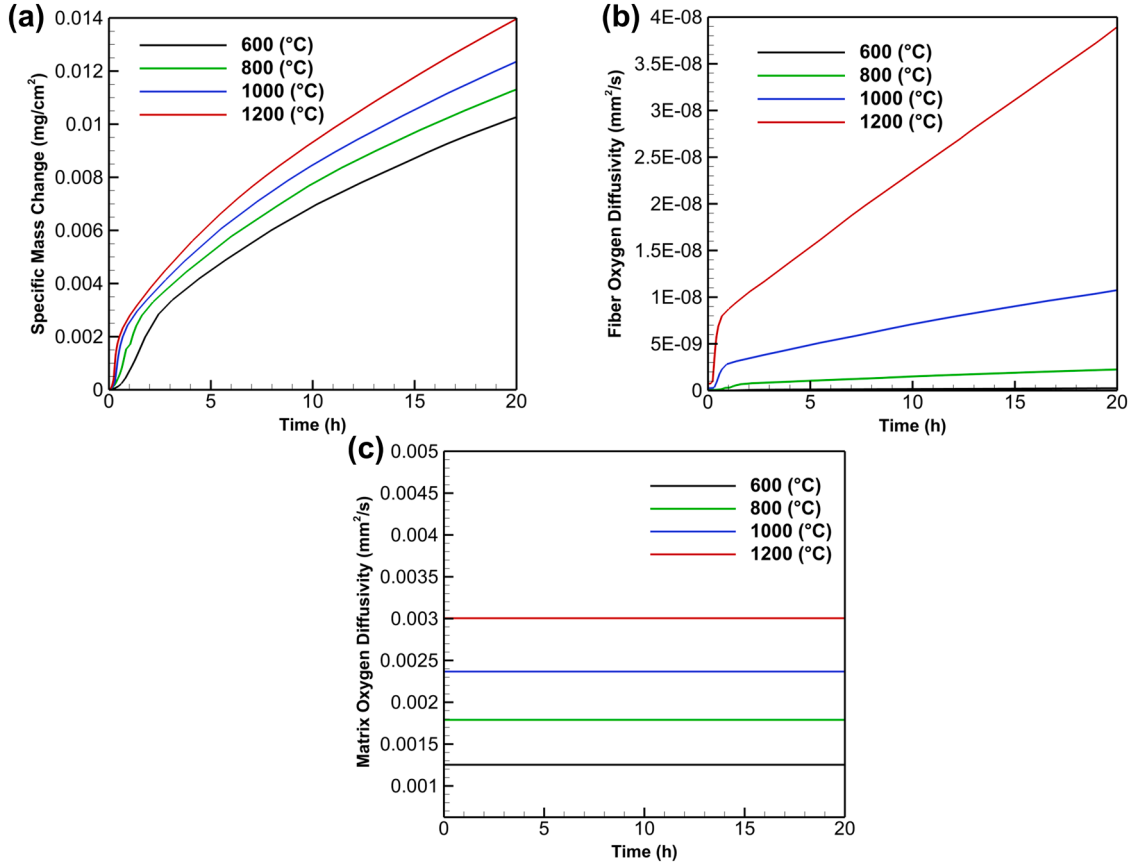


Fig. 2. Averaged unstressed oxidation results for SiC/SiC SRVE at intermediate and high temperatures; (a) Specific mass change due to silica formation; (b) Oxygen diffusivity evolution in fiber; (c) Oxygen diffusivity evolution in matrix.

brick elements.

The Galerkin method of weighted residuals is used to derive the weak forms of the equilibrium and conservation of mass equations (Eqs. (1) and (4)) with natural boundary conditions. These equations are discretized and reduced to the elemental form as follows.

$$\mathbf{F}_u^e = \int_{\Omega^e} \mathbf{B}^T \boldsymbol{\sigma}^e d\Omega^e - \int_{\Gamma^e} \mathbf{N}^T \bar{\mathbf{t}} d\Gamma^e \quad (12a)$$

$$\mathbf{F}_{\phi_{O_2}}^e = \int_{\Omega^e} \left(\begin{array}{c} \mathbf{N}^T \mathbf{N} \phi_{O_2}^e \frac{d\phi_{O_2}}{dt} \\ + \mathbf{N}^T \mathbf{N} s_{O_2} \frac{d\phi_{O_2}}{dt} \\ + \nabla \mathbf{N}^T \nabla \mathbf{N} D_{O_2} s_{O_2} \phi_{O_2}^e \\ + 3\mathbf{N}^T (k K_f s_{O_2} s_{SiO_2} \mathbf{N} \phi_{O_2}^e (1 - \mathbf{N} \phi_{SiO_2}^e)) \end{array} \right) d\Omega^e + \int_{\Gamma^e} \mathbf{N}^T \bar{J}_{O_2} d\Gamma^e \quad (12b)$$

$$\mathbf{F}_{\phi_{SiO_2}}^e = \int_{\Omega^e} \left(\begin{array}{c} \mathbf{N}^T \mathbf{N} s_{SiO_2} \frac{d\phi_{SiO_2}}{dt} \\ - 2\mathbf{N}^T (k K_f s_{O_2} s_{SiO_2} \mathbf{N} \phi_{O_2}^e (1 - \mathbf{N} \phi_{SiO_2}^e)) \end{array} \right) d\Omega^e \quad (12c)$$

where $\mathbf{F}_u^e$, $\mathbf{F}_{\phi_{O_2}}^e$, and $\mathbf{F}_{\phi_{SiO_2}}^e$ are the elemental residual forces corresponding to the equilibrium equations, and the conservation of mass for oxygen and silica, respectively. Ω^e and Γ^e denote the volume and natural boundary of the element domain, respectively. $\mathbf{B}$ is the symmetric discrete gradient matrix, $\mathbf{N}$ is the shape function matrix for isoparametric elements, and $\mathbf{N}$ is the shape function matrix scattered for three displacement degrees of freedom. $\bar{\mathbf{t}}$ is the applied traction vector

and $\bar{J}_{O_2}$ is the prescribed oxygen flux on the natural boundary of the element. Eq. (12) is applied to each CMC microstructure constituent, while Eqs. (9)–(11) are used to compute the oxygen solubility and diffusivity of the corresponding constituent. The integration over the element domain is performed numerically using Gaussian quadrature for fully integrated linear elements.

The backward-Euler scheme and standard Newton-Raphson method in Abaqus, featuring automatic time step adjustment based on problem nonlinearity and automatic convergence checks applied to the residuals, are used to solve the nonlinear problem. Under applied loading, rapid oxidation of SiC fibers is initially observed, resulting in primary creep-like strain, which requires relatively small time steps (on the order of fractions of a second). However, as silica forms on the fiber surface, the oxidation rate decreases, leading to fiber sealing and allowing for larger time steps (up to several hours). The nonlinear differential equations are linearized within each time step, with quantities like solubility and diffusivity remaining constant within a time step but updated with every iteration. The local elemental residual forces (Eq. (12)) are converted to the following matrix form:

$$\begin{bmatrix} \mathbf{F}_u^e \\ \mathbf{F}_{\phi_{O_2}}^e \\ \mathbf{F}_{\phi_{SiO_2}}^e \end{bmatrix} = - \begin{bmatrix} \mathbf{K}_{uu}^e & \mathbf{K}_{u\phi_{O_2}}^e & \mathbf{K}_{u\phi_{SiO_2}}^e \\ \mathbf{K}_{\phi_{O_2}u}^e & \mathbf{K}_{\phi_{O_2}\phi_{O_2}}^e & \mathbf{K}_{\phi_{O_2}\phi_{SiO_2}}^e \\ \mathbf{K}_{\phi_{SiO_2}u}^e & \mathbf{K}_{\phi_{SiO_2}\phi_{O_2}}^e & \mathbf{K}_{\phi_{SiO_2}\phi_{SiO_2}}^e \end{bmatrix} \begin{bmatrix} d\mathbf{u}^e \\ d\phi_{O_2}^e \\ d\phi_{SiO_2}^e \end{bmatrix} \quad (13)$$

where $d\mathbf{u}^e$, $d\phi_{O_2}^e$, and $d\phi_{SiO_2}^e$ are the incremental element nodal displacements, and normalized concentrations of oxygen and silica, respectively. $\mathbf{K}^e$ is the elemental Jacobian matrix which is the total derivative of the residual force vector with respect to each degree of freedom (i.e., $\mathbf{K}_{u\phi_{O_2}}^e = -\frac{d\mathbf{F}_u^e}{d\phi_{O_2}^e}$). Note that the residual forces are used to

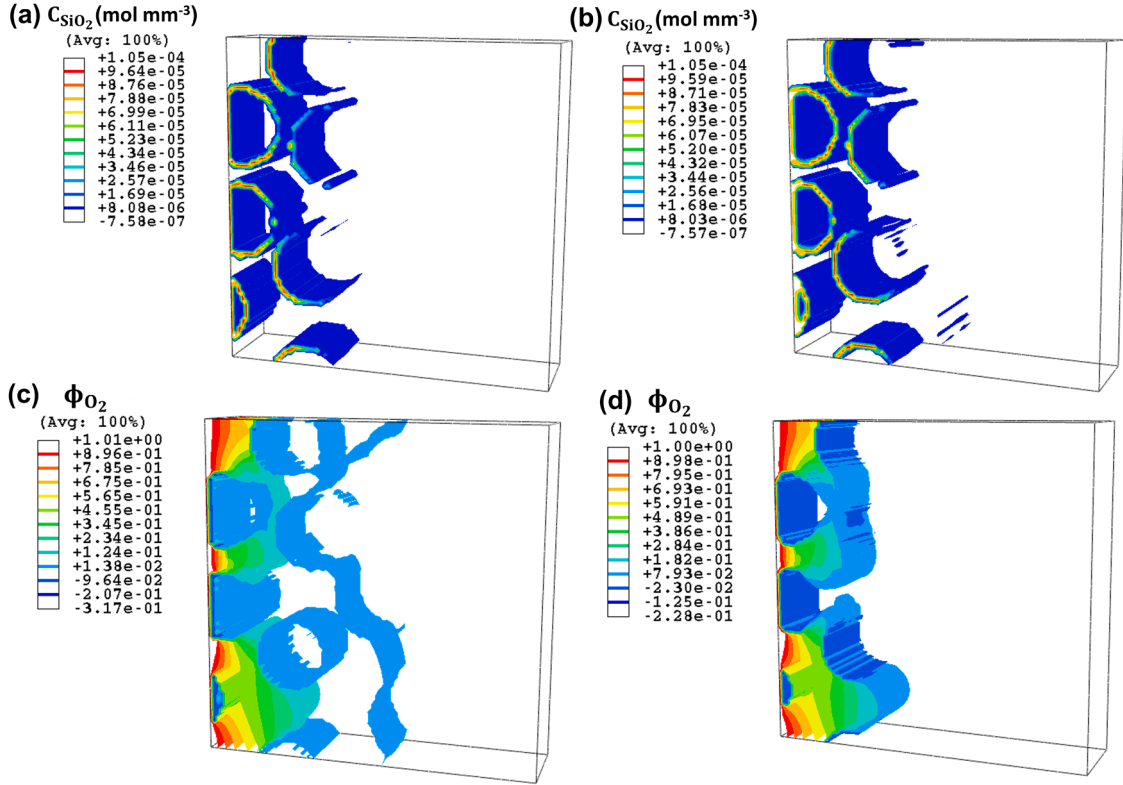


Fig. 3. Unstressed oxidation localized results in terms of silica concentration (C_{SiO_2}) and normalized oxygen concentration (ϕ_{O_2}) after 20 h of air exposure. (a) and (c) show the distribution of C_{SiO_2} and ϕ_{O_2} , respectively, at 600 °C, while (b) and (d) present the results at 1200 °C.

ensure solution accuracy, and the complete definition of the elemental Jacobian is not necessary (i.e., terms with intractable tangent modulus, in case of complex damage model, can be approximated). However, the Jacobian aids in convergence, and in this work, the full definition of the Jacobian matrix, including both the direct and indirect dependencies of the degrees of freedom on the residual forces, is utilized. This is due to the high degree of nonlinearity in the multiphysics model and the highly heterogeneous CMC microstructure with regions of localized deformation. The full Jacobian is provided in the Appendix. Finally, K^e can be decomposed into elemental stiffness and capacitance matrices [23], where the capacitance matrix can be diagonalized for linear elements to reduce numerical oscillations. However, the choice of such diagonalization scheme highly depends on the nonlinearity and the numerical stability of the problem.

3.2. Simulation setup

A recently developed DL based SRVE generation algorithm [36–38] is used to generate realistic CMC microstructures, accounting for stochastic distributions of the intratow fiber and porosity volume fractions as shown in Fig. 1a. The SRVEs consist of three types of material sub-cells: SiC fiber, SiC matrix, and porosity. Randomly distributed unidirectional fibers are allocated within the SRVE, where the fibers at the SRVE edges are continuous since the ergodic hypothesis is invoked [38, 55]. Porosity is distributed within the matrix, specifically around the fiber/matrix interface, based on intratow microstructure characterization. The fiber and porosity volume fractions utilized in this study are 49 % and 3 %, respectively. Periodic boundary conditions (PBCs) are applied along X and Z directions, while free surface boundary condition is applied to the transverse fiber direction (Y-direction) to allow oxygen to diffuse within the SRVE [23,56]. The node at the origin is fixed to eliminate rigid body motion. The elastic material properties for the SiC matrix, Hi-Nicalon™ fibers (SiC fibers), and silica (SiO_2) are obtained from literature [22,38,41] and are shown in Table 1, where E and ν are

the isotropic Young's modulus, and Poisson's ratio respectively, and σ_{UTS} is the ultimate tensile strength which is needed for the matrix damage model [34,42]. The multiphysics model parameters with regards to solubility, diffusivity, and chemical reaction activation energies and rate constants are summarized in Table 2.

Unstressed oxidation simulations are conducted to mimic the TGA tests, where each of the pre-damaged SRVEs, with only manufacturing-induced defects, is simulated at a constant temperature, and the change in mass due to silica formation is recorded. This provides insight into the oxidation regime and associated oxidation kinetics. Stressed oxidation simulations are conducted to analyze the effect of damage evolution on the oxidation kinetics. The SRVE is subjected to in-plane creep deformation with stress free boundary conditions applied in all other directions to enable Poisson's effect. In particular, the load on the SRVE is ramped to a certain stress level in the axial fiber direction over a period of 10 s and then held constant for the rest of the simulation time as shown in Fig. 1b. Each stressed oxidation simulation is repeated at stress levels ranging from 70 MPa to 200 MPa to investigate the effect of matrix damage on the oxidation kinetics. The applied stress level range is selected such that the matrix stress can exceed the microcracking stress, thereby ensuring sufficient matrix damage to provide further oxygen diffusive channels and infiltration to the SiC fibers. For this study, both unstressed and stressed oxidation simulations are conducted at constant temperatures ranging from 600 °C to 1200 °C. Note that since there is no temperature ramping in the oxidation simulations, the effect of constituents' thermal mismatch on the thermomechanical response and oxidation kinetics are not considered. While several studies in the literature focus on the oxidation of SiC/SiC CMCs at high temperatures (1200–1400 °C) [6,20,26,57], only a few studies have examined the oxidation of SiC/SiC and SiC fibers at intermediate temperatures (500–800 °C). Detwiler et al. [18] studied the oxidation of CVI SiC/SiC at 600 °C and 800 °C in dry and wet oxidizing environments. Their findings indicated that the onset of oxidation occurred between 450–480 °C in a dry oxygen environment, whereas in a wet oxygen

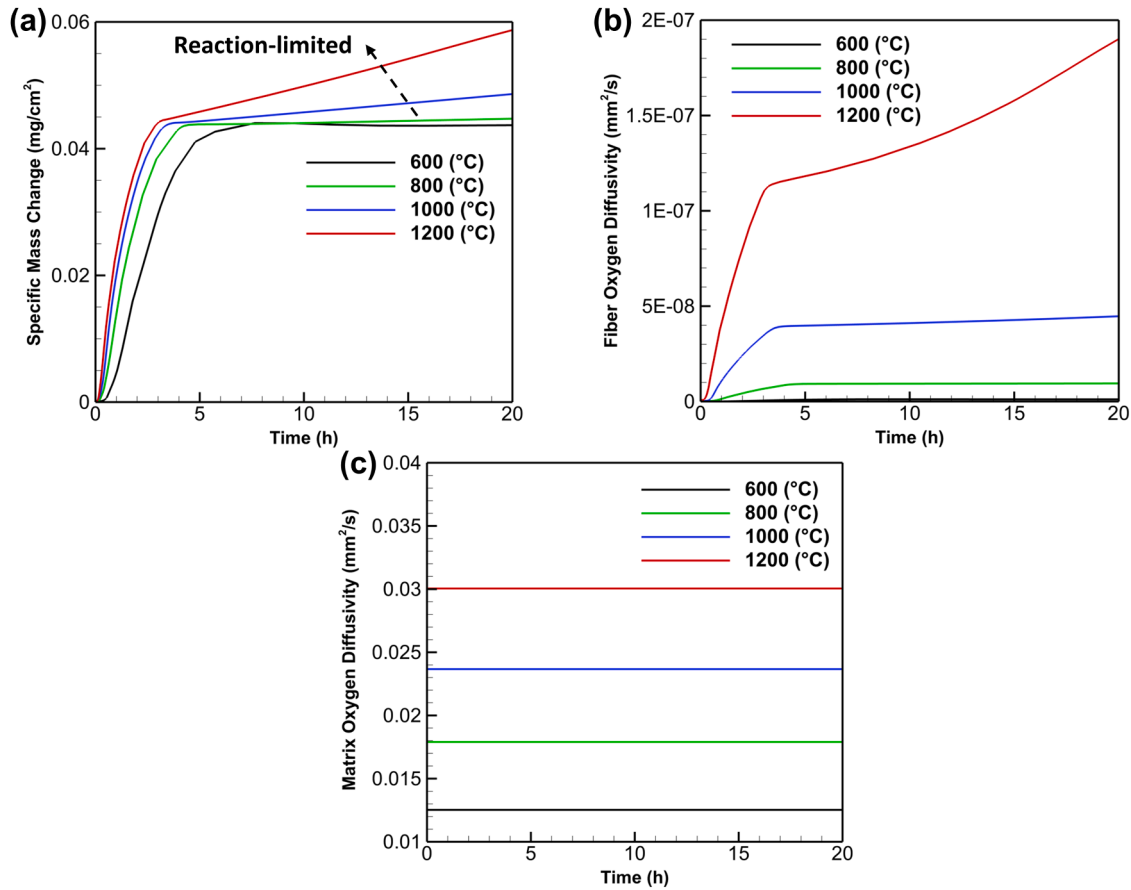


Fig. 4. Averaged unstressed oxidation results for SiC/SiC SRVE with higher initial matrix defects (case 2) at intermediate and high temperatures; (a) Specific mass change due to silica formation; (b) Oxygen diffusivity evolution in fiber; (c) Oxygen diffusivity evolution in matrix.

environment, oxidation began at 450 °C. Furthermore, Gauthier et al. [21] investigated the oxidation of various SiC fibers, including Hi-Nicalon™ fibers, during static fatigue at intermediate temperatures, where the formation of a thin silica film on the fiber surface was evident. Gogotsi et al. [58] demonstrated that SiC fiber oxidation begins at 600 °C, while Wilson et al. [5] also studied the oxidation of SiC fibers at temperatures below 800 °C. The intermediate temperature range is of particular interest, as the silica formation rate at these temperatures is too slow to provide effective fiber sealing against oxygen ingress. Consequently, oxidative degradation mechanisms are more detrimental to the thermomechanical performance of SiC/SiC CMCs within this temperature range.

4. Results and discussion

The oxidation kinetics and the silica scale thickness growth are investigated for unstressed and stressed oxidation simulations at various temperatures. The unstressed oxidation simulations are conducted with two microstructures, with different initial matrix states, defined by the parameter α_2 in Table 2, to study the impact of the available oxygen diffusive channels on the unstressed oxidation regimes. Case 1 ($\alpha_2 = 1 \times 10^{-3}$) represents a microstructure with fewer matrix defects, and case 2 ($\alpha_2 = 1 \times 10^{-2}$) represents a microstructure with more matrix defects. Note that, in both cases, pores are distributed around the fiber surfaces; hence, an increase in α_2 value would mimic additional available pores or matrix microcracks.

It must be noted that a quantitative validation of CMC oxidation models is challenging due to the differences in length scales between the microscale simulations and the TGA/stress rupture experiments. PIP SiC/SiC mini-composite oxidation data is the closest to the SRVE

simulation data; however, there is limited availability of such experimental data in the literature due to the high cost of the material and proprietary restrictions on public and academic release. Therefore, qualitative validation of both unstressed and stressed oxidation results is provided.

4.1. Unstressed oxidation

Fig. 2 shows the results of unstressed oxidation simulations conducted for microstructure case 1. The specific mass change¹ due to silica formation at the fiber surfaces is computed for various temperatures as shown in Fig. 2a. Mass gain is observed at all temperatures, and the silica scale growth indicates parabolic oxidation kinetics. This agrees with results in the literature for Hi-Nicalon fiber and SiC/SiC oxidation [5, 20]. The average diffusivity of oxygen in fiber increases with time due to the silica growth, as shown in Fig. 2b, since oxygen diffusivity in silica is significantly higher than in SiC fibers, as mentioned in Section 2.2. The mesh elements of the SiC fiber transform into silica due to the passive oxidation chemical reaction, and hence the diffusivity of oxygen in fiber increases. The average oxygen diffusivity in matrix is shown in Fig. 2c, where it increases with temperature according to Eq. (11d). The oxygen diffusive channels in the matrix remain unchanged since there is no mechanical loading-induced damage, and the oxidation of the PIP matrix is ignored. This results in constant oxygen diffusivity in matrix

¹ The specific mass change (Δm) depends on the molar masses of silica and SiC fiber, denoted as M_{SiO_2} and M_{SiC} , respectively. $\Delta m = \frac{\bar{C}_{SiO_2} \times (M_{SiO_2} - M_{SiC}) \times V_{SiO_2}}{A_f}$, where $\bar{C}_{SiO_2}$ is the average silica concentration in the SRVE, V_{SiO_2} is the volume of the mesh elements which have silica, and A_f is the fibers surface area.

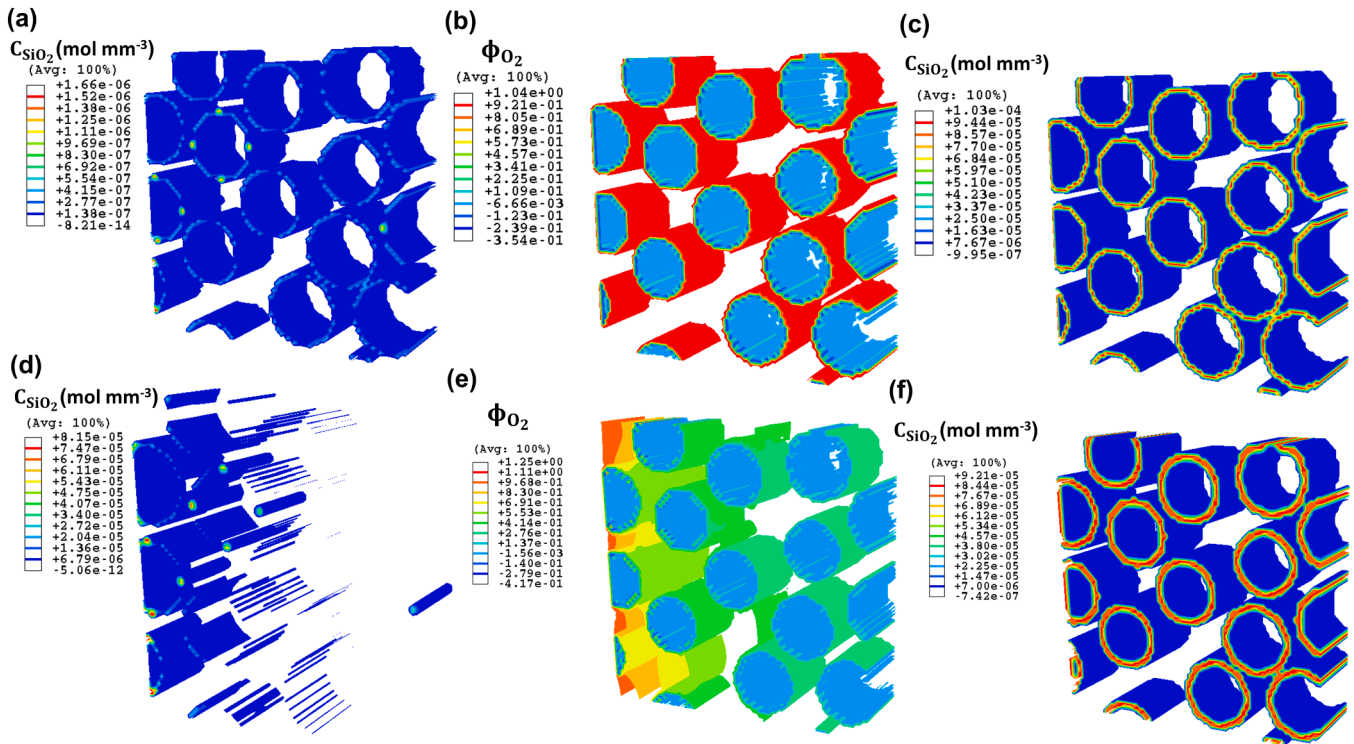


Fig. 5. Unstressed oxidation localized results in terms of silica concentration (C_{SiO_2}) and normalized oxygen concentration (ϕ_{O_2}) for SRVE with higher initial matrix defects (case 2) after air exposure. At 600 °C, (a) and (b) show the distribution of C_{SiO_2} and ϕ_{O_2} , respectively, after 16 min, while (c) shows the silica saturated thickness after 20 h. Similarly, at 1200 °C, (d) and (e) present the results after 16 min, and (f) shows the silica thickness as a result of reaction-limited oxidation regime shift after 20 h.

throughout the simulation time. The spatial distributions of silica concentration and normalized oxygen concentration after 20 h of air exposure are shown in Fig. 3. The fibers near the surface, where the oxygen flux is defined, oxidize first as oxygen infiltrates the SRVE. At 600 °C, Fig. 3a shows elements with high C_{SiO_2} near the surface, gradually decreasing throughout the SRVE. This indicates that the oxidation process is diffusion-limited, where the chemical reaction rate is faster than the oxygen diffusivity in matrix, and the fibers oxidize as soon as their surfaces interact with oxygen. At sufficient distance from the oxygen flux surface, there is no oxidation, and more exposure time is needed to fully oxidize the remaining fibers. Fig. 3b shows the silica concentration results at 1200 °C. The oxidation is still diffusion-limited; however, compared to 600 °C case, higher silica concentrations are observed, and oxygen infiltrates further into the SRVE, oxidizing a greater surface area of the fibers. Fig. 3c and d compare the normalized oxygen concentration within the SRVE at 600 °C and 1200 °C, respectively. In the diffusion-limited oxidation regime, the oxidation of the silica near the exposed SRVE surface provides a shielding effect to the interior fibers, as the lack of diffusive channels in the matrix allows oxygen to become trapped within the produced silica. This is evident in Fig. 3d, where oxygen is clustered near the exposed surface. However, at 600 °C, the silica concentration is lower, allowing oxygen to penetrate further into the microstructure through the limited diffusive channels. The diffusion-limited oxidation, resulting in the evolution of silica within the SRVE at 1200 °C, is shown in Supplementary Movie S1.

The results from the microstructure with more matrix defects (case 2) are presented in Fig. 4. The specific mass change is shown in Fig. 4a, indicating that the mass gain follows parabolic oxidation kinetics at temperatures lower than 1000 °C. Due to the abundance of oxygen diffusive channels, a critical silica thickness is reached at the fiber surfaces which seals the SiC fibers beneath the silica scale from further oxidation. At higher temperatures (≥ 1000 °C), the oxidation kinetics are para-linear. Initially, the oxidation is diffusion-limited, however as the

silica thickness grows, the kinetics become linear, transitioning to a reaction-limited oxidation regime. The transformation from diffusion-limited to reaction-limited oxidation has been observed in the literature for Si nanowires, SiC fibers, chemical vapor deposition (CVD) SiC thick films, and SiC/SiC CMCs [18,59,60]. The average oxygen diffusivity in fibers, as shown in Fig. 4b, exhibits the same trend as the silica mass change. The diffusivity increases at a different rate once the reaction-limited oxidation regime is reached, rather than saturating, and this change in diffusivity is amplified at 1200 °C. The average diffusivity of oxygen in matrix is shown in Fig. 4c, where it exhibits the same behavior as in the case 1 simulations. However, the diffusivity levels are higher by one order of magnitude due to the increased matrix defects.

The spatial distributions of silica concentration and normalized oxygen concentration for microstructure case 2 after air exposure at 600 °C and 1200 °C are shown in Fig. 5. After 16 min of exposure at 600 °C, silica forms homogeneously across the SRVE due to the abundance of oxygen diffusive channels, as shown in Fig. 5a. Consequently, the normalized oxygen concentration is equally distributed among all the fiber surfaces, as shown in Fig. 5b. After 20 h of air exposure at 600 °C, significant oxidation is observed throughout the fibers, as shown in Fig. 5c, although oxygen could not radially infiltrate further into the fibers, and the silica thickness saturated. After 16 min of exposure at 1200 °C, an inhomogeneous silica distribution is observed (Fig. 5d), with higher silica concentrations near the oxygen flux surface. This is attributed to the higher chemical reaction rate at 1200 °C, where fibers oxidize to form silica and subsequently shield the interior fibers, like in Fig. 3d. The normalized oxygen distribution follows the same trend, where oxygen is more concentrated at the fibers near the exposed SRVE surface, as shown in Fig. 5e. After 20 h of exposure at 1200 °C, greater final silica thickness is observed due to the shift from diffusion-limited to reaction-limited oxidation, as shown in Fig. 5f. The full silica evolution with oxidation regime shift is shown in Supplementary Movie S2.

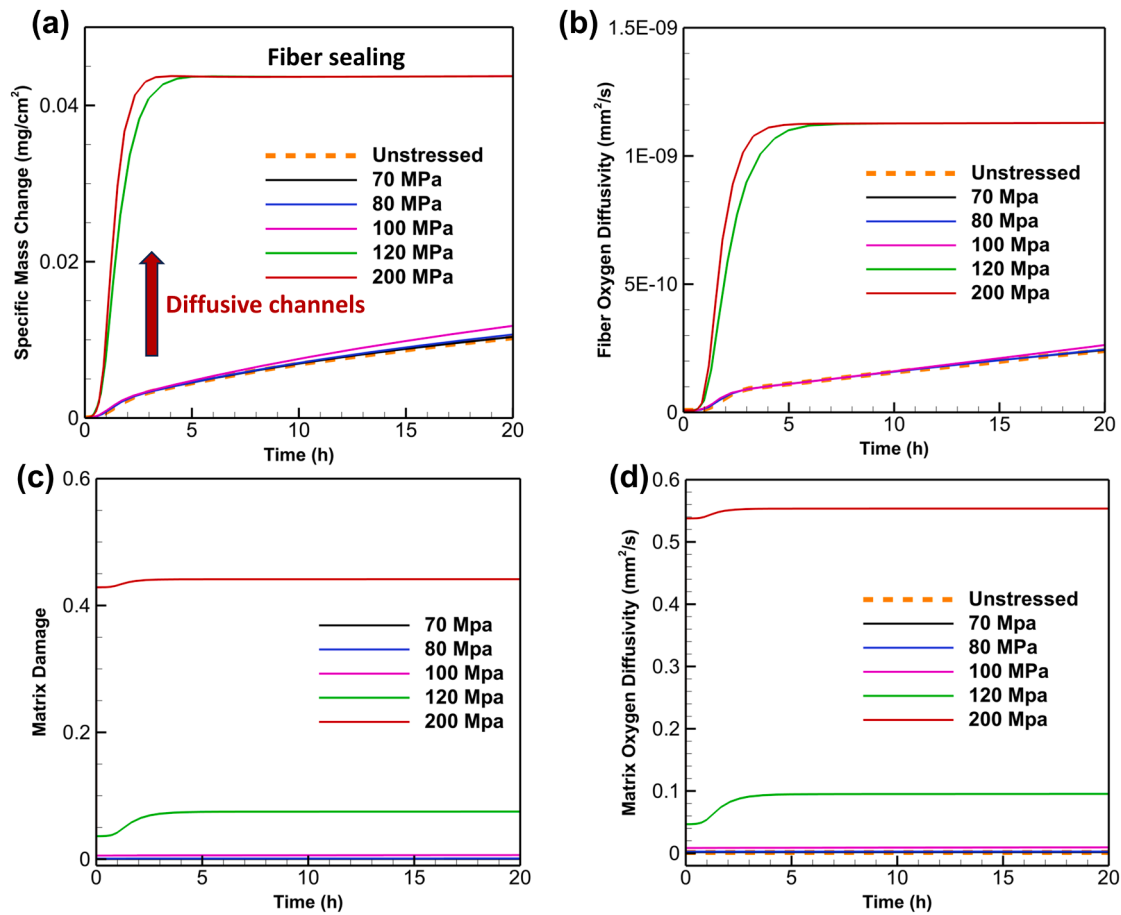


Fig. 6. Averaged stressed oxidation results for SiC/SiC SRVE at 600 °C (intermediate temperature); (a) Specific mass change due to silica formation; (b) Oxygen diffusivity evolution in fiber; (c) Matrix damage evolution due to applied loading and fiber/matrix load transfer; (d) Oxygen diffusivity evolution in matrix.

4.2. Stressed oxidation

To investigate the effect of microcrack evolution on oxidation kinetics and the time-dependent response, the SRVE in case 1 is subjected to uniaxial creep loading at various stress levels. The simulations are conducted at 600 °C to examine fiber sealing behavior and at 1200 °C to investigate reaction-limited stressed oxidation. Note that case 1, with low initial matrix damage, is selected to study the formation and evolution of matrix damage during the loading ramp-up (shown in Fig. 1). The stressed oxidation results at 600 °C are compared with the unstressed case in Fig. 6. The trends of the specific mass change for applied loading up to 100 MPa are similar to the unstressed simulation as shown in Fig. 6a. However, once the load reaches 120 MPa, significant silica formation and mass change are observed, thereafter the mass change saturates, indicating that the critical silica scale thickness at this temperature has been reached resulting in fiber sealing. The average fiber oxygen diffusivity exhibits trends similar to the specific mass change as shown in Fig. 6b, which is consistent with the unstressed oxidation results. The significant increase in the silica formation at 120 MPa and above is due to the average matrix damage evolution as shown in Fig. 6c. At these high stress levels, the matrix stress exceeds the microcrack stress and results in an increase in the oxygen diffusive channels. Note that our damage model is a softening damage law, as discussed in Section 2.1. The damage parameter is linked to matrix oxygen diffusivity and solubility (Eqs. (9a) and (11a)), mimicking the effect of matrix microcrack evolution. The corresponding average oxygen diffusivity in the matrix is shown in Fig. 6d, where an increase in matrix diffusivity with increasing applied load levels is observed. Contrary to the unstressed oxidation case, the matrix oxygen diffusivity changes during the

creep simulation hold time. This is attributed to fiber/matrix load transfer due to oxidation. During oxidation, the fiber mesh elements transform into silica which has lower stiffness compared to SiC, consequently the average fiber stiffness drops and the fiber sheds load to the matrix to maintain the same composite creep stress level. Depending on the degree of fiber oxidation, the increase in matrix stress increases the number of matrix microcracks and oxygen diffusivity during the hold time.

The stressed oxidation results at 1200 °C are summarized in Fig. 7. The oxidation kinetics are similar to the 600 °C simulations for applied stress levels below 120 MPa, with both exhibiting diffusion-limited oxidation regime. However, when the applied stress exceeds the matrix microcrack stress, the specific mass change exhibits para-linear oxidation kinetics, as shown in Fig. 7a. The para-linear behavior at elevated stresses is similar to the unstressed simulation results (case 2) shown in Fig. 4; however, the applied loading induces further matrix defects, and the transition from diffusion-limited to reaction-limited oxidation occurs at a faster rate. The average fiber oxygen diffusivity behavior as shown in Fig. 7b is consistent with the previous discussion (i. e., follows the trend of silica mass change). The oxygen diffusivity in fiber at 1200 °C is two orders of magnitude higher compared to that at 600 °C. This increase is attributed to the dependence of diffusivity on both temperature and silica scale concentration, as discussed in Eq. (11b). Hence, sufficient oxygen for the chemical reaction can infiltrate into the SiC fibers below the silica scale, where the fibers exhibit reaction-limited oxidation. Fig. 7c and d present the evolution of matrix damage and oxygen diffusivity, respectively, and the trends are similar and agree with our previous discussions. The spatial distributions of the silica and oxygen concentration parameters, and damage in the SRVE

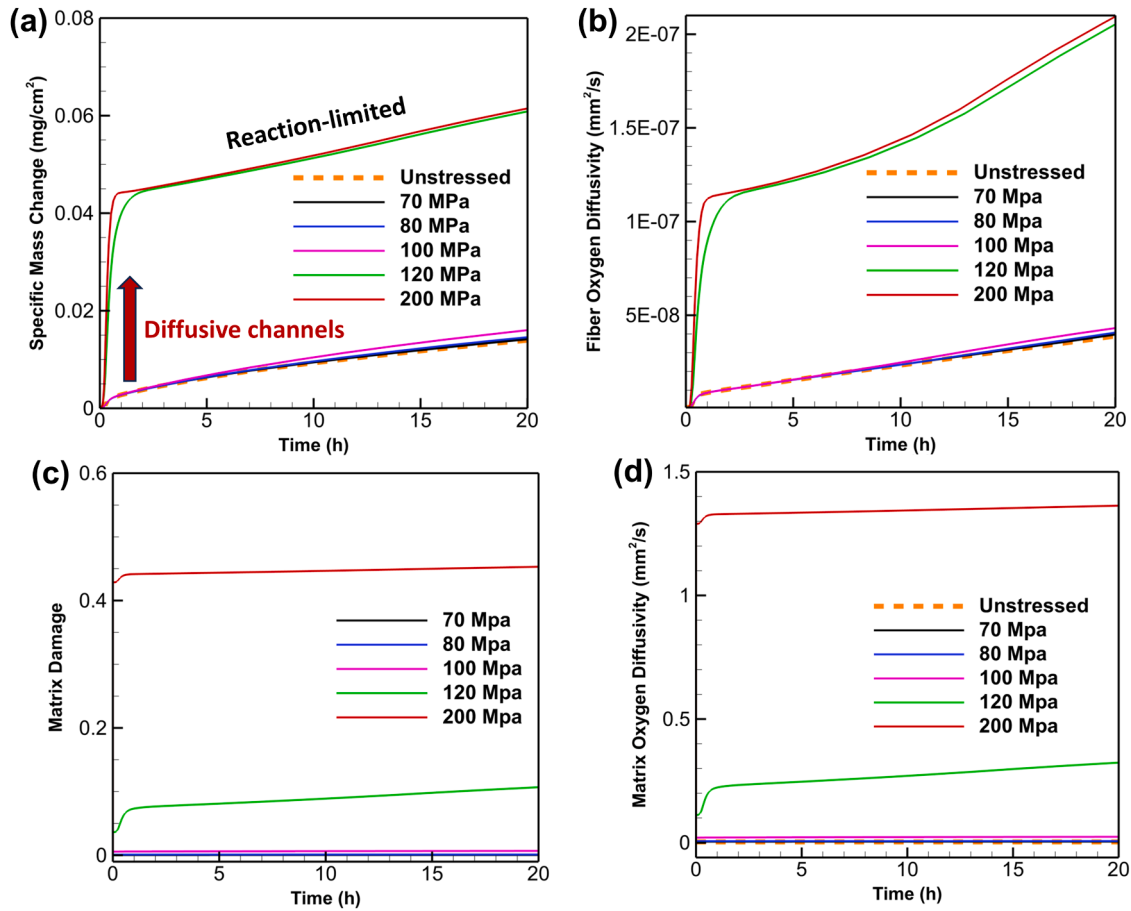


Fig. 7. Averaged stressed oxidation results for SiC/SiC SRVE at 1200 °C (high temperature); (a) Specific mass change due to silica formation; (b) Oxygen diffusivity evolution in fiber; (c) Oxygen diffusivity evolution in matrix; (d) Matrix damage evolution due to applied loading and fiber/ matrix load transfer.

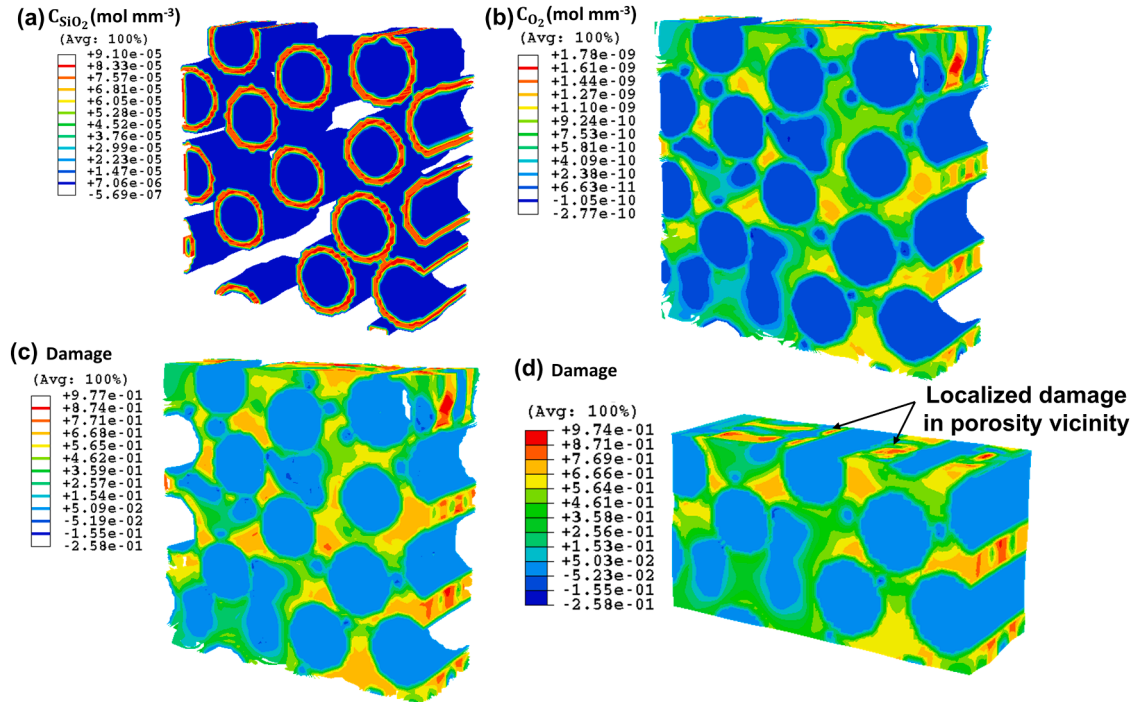


Fig. 8. Stressed oxidation localized results after 20 h of air exposure at 1200 °C, with a loading of 200 MPa applied in the axial fiber direction. (a) Silica concentration (C_{SiO_2}); (b) Oxygen concentration (C_{O_2}); (c) Damage in the SRVE front view; (d) Damage in an SRVE section, highlighting the interior porosity-damage interaction.

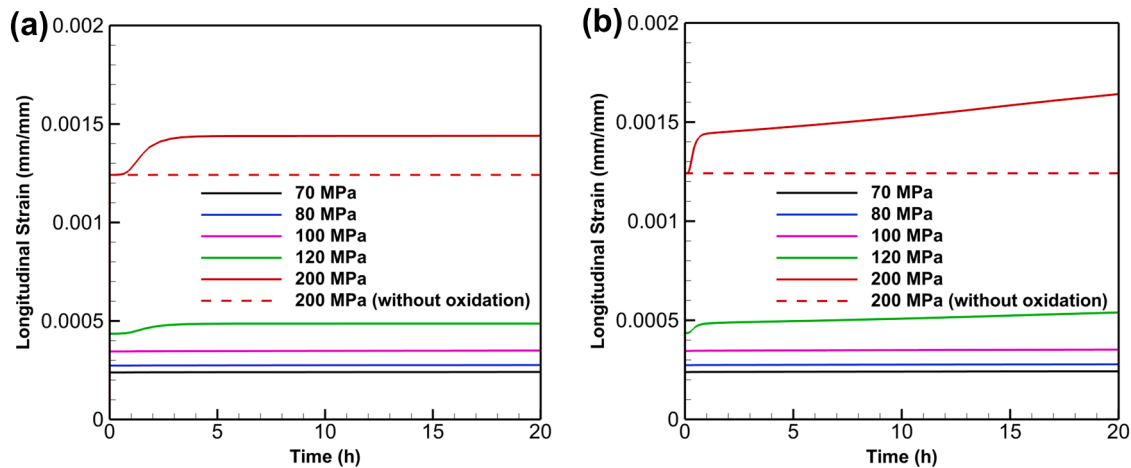


Fig. 9. Creep-like strain response of SiC/SiC SRVE at various uniaxial creep load levels for (a) 600 °C and (b) 1200 °C.

subjected to creep loading of 200 MPa for 20 h are shown in Fig. 8. The formation of silica concentration is homogeneous at all the fiber surfaces, which agree with the results from reaction-limited dominant oxidation regime shown in Fig. 7a. Additionally, the oxygen concentration shown in Fig. 8b follows the matrix damage locations (Fig. 8c) in the SRVE which confirms the average matrix diffusivity results. A section of the SRVE is presented in Fig. 8d to show the damage distribution along the fiber direction and pores. Porosity acts as a hot spot and induces localized deformation and matrix damage. This creates diffusive pathways for oxygen along the fiber surfaces, maintaining oxygen exposure during the reaction-limited oxidation regime. The formation of damage and diffusive channels near the fiber surface because of oxidation agrees with the literature results [13,61]. Silica and damage evolution in the SRVE at 200 MPa and 1200 °C are shown in Supplementary Movies S3 and S4, respectively.

The creep-like strain behavior induced by oxidation damage is analyzed next, as shown in Fig. 9. Fiber damage due to silica formation leads to fiber/matrix load transfer, similar to what is observed in stress rupture experiments in inert environments. At high temperatures, CMCs experience primary (Coble creep) and secondary (Nabarro-Herring creep) creep [26]. The Coble creep in Hi-Nicalon fiber is dominated by viscoelastic deformation of carbon at the grain boundaries (GBs), leading to GB sliding [62]. Nabarro-Herring creep is driven by diffusion of impurities along the GBs, which results in further GB sliding and lattice bulk diffusion [16,62]. In addition to these thermally activated processes, oxidation also contributes to the creep behavior. The longitudinal strain along the fiber direction for the 600 °C creep simulations is shown in Fig. 9a. At stress levels below 120 MPa, the microstructure does not exhibit any creep deformation, since the fiber stiffness degradation and constituents' load transfer are not significant due to the low silica scale formation. However, at higher stress levels, more silica scale formation is observed which accelerates the creep-like strain. The diffusion-limited oxidation results in primary creep-like strain. However, once all the fibers are sealed, the constituents' load transfer is halted, leading to constant strain for the rest of the hold time. At 1200 °C and at stress levels above 120 MPa, the creep-like strain is observed as shown in Fig. 9b. The inelastic strain exhibits the primary and secondary creep-like strain stages. The diffusion-limited mechanism is responsible for the primary creep-like strain due to its parabolic oxidation kinetics, while the shift to the reaction-limited oxidation regime results in secondary creep-like strain due to its linear oxidation kinetics. Finally, the chemical reaction activation energy and rate constants are increased to arbitrary high values to simulate the thermomechanical response in inert atmosphere (damage without oxidation), where the CMC microstructure does not exhibit any inelasticity. These inert atmosphere simulations are accurate at 600 °C; however, at 1200 °C, the thermally

activated microstructure processes dominate the inelastic deformation [38,63]. Hence, the coupling of a viscoplastic creep model with the developed multiphysics oxidation model would be capable of capturing both primary and secondary creep due to oxidation damage and thermally activated microstructural mechanisms.

5. Conclusion

A multiphysics model was developed to simulate the oxidation kinetics of SiC/SiC CMCs at elevated temperatures, explicitly modeling oxygen diffusion through SRVEs, which are high-fidelity representations of material microstructures with manufacturing-induced defects. The governing equations for equilibrium and conservation of mass were linked through SiC fiber oxidation reaction rates, matrix damage, and oxygen solubility and diffusivity within the CMC interior. The weak forms of the governing equations were derived using the Galerkin method of weighted residuals and then discretized for numerical implementation using FEA to simulate both unstressed and stressed oxidation tests. The model captures the diffusion-limited oxidation regime during unstressed simulations of microstructure with low initial matrix defects, where at high temperatures, the oxidation of fibers near the oxygen flux surface shields the interior fibers against oxidation. Unstressed oxidation simulations with high initial matrix defects show that the oxidation regime is diffusion-limited with silica scale saturation at temperatures below 1000 °C. However, above this temperature, the oxidation regime shifts to reaction-limited due to the high oxygen diffusivity in the silica scale. Additionally, the model captures the stressed oxidation behavior, where applied creep stress levels above 120 MPa increases the number of oxygen diffusive channels in the matrix and near the fiber surfaces, resulting in faster oxidation rates. The strain acceleration behavior of stressed oxidation is also captured, where SiC fiber oxidation leads to fiber/matrix load transfer. The diffusion-limited oxidation regime contributes to the primary creep-like strain, while the reaction-limited oxidation regime contributes to the secondary creep-like strain. The results clearly illustrate the importance and usefulness of the developed high-fidelity multiphysics model in identifying the fundamental mechanisms that lead to damage and material degradation in CMCs operating in a thermomechanical oxidative environment.

Future work will extend the multiphysics model to incorporate the effects of volumetric changes and the resulting induced stresses associated with the phase transformation of SiC to silica scale on oxidation kinetics and oxidative degradation mechanisms. Furthermore, both the fiber and matrix exhibit creep behavior (viscoplasticity) in vacuum due to grain boundary sliding and bulk lattice diffusion. Therefore, a strain-hardening three-dimensional viscoplastic model should be coupled with

the current multiphysics model to capture the inelastic response resulting from both oxidation-induced damage and thermally activated microstructural mechanisms.

CRedit authorship contribution statement

Mohamed H. Hamza: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jacob J. Schichtel:** Software, Methodology, Investigation, Conceptualization. **Aditi Chattopadhyay:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

To implement the model in Abaqus UEL, the full model equations are provided. The elemental residual forces and Jacobian for each microstructure constituent- fiber (f), matrix (m), and porosity- depend on oxygen solubility and diffusivity, chemical reaction activation energy and rate constant, as well as the matrix damage parameter. In this context, the equations are presented with the boundary condition term omitted, as the boundary conditions are defined at the microstructure nodes and are incorporated by the Abaqus solver. In the derivation, the following parameters are defined

as: $R_c = k k_0 \exp\left(\frac{-Q_c}{RT}\right)$; $\Delta s = s_{\max O_2} - s_{\min O_2}$; $\Delta D = D_{\max O_2} - D_{\min O_2}$; $D^0 = D_{O_2}^0 \exp\left(\frac{-Q_{O_2}}{RT}\right)$.

The equations of the SiC fiber are as follows.

$$\mathbf{F}_{\phi_{O_2}}^{e(F)} = \int_{\Omega^e} \left(\begin{array}{c} \mathbf{N}^T \mathbf{N} \Phi_{O_2}^e \mathbf{N} \frac{d\Phi_{SiO_2}^e}{dt} \Delta s^F \\ + \mathbf{N}^T \mathbf{N} (\Delta s^F \mathbf{N} \Phi_{SiO_2}^e + s_{\min O_2}^F) \frac{d\Phi_{O_2}^e}{dt} \\ + \nabla \mathbf{N}^T \nabla \mathbf{N} \Phi_{O_2}^e \left(D^0 (\mathbf{N} \Phi_{SiO_2}^e)^2 + D_{\min O_2}^F \right) (\Delta s^F \mathbf{N} \Phi_{SiO_2}^e + s_{\min O_2}^F) \\ + 3 \mathbf{N}^T (R_c s_{SiO_2} (\Delta s^F \mathbf{N} \Phi_{SiO_2}^e + s_{\min O_2}^F) \mathbf{N} \Phi_{O_2}^e (1 - \mathbf{N} \Phi_{SiO_2}^e)) \end{array} \right) d\Omega^e \quad (A1)$$

$$\mathbf{F}_{\phi_{SiO_2}}^{e(F)} = \int_{\Omega^e} \left(\begin{array}{c} \mathbf{N}^T \mathbf{N} s_{SiO_2} \frac{d\Phi_{SiO_2}^e}{dt} \\ - 2 \mathbf{N}^T (R_c s_{SiO_2} (\Delta s^F \mathbf{N} \Phi_{SiO_2}^e + s_{\min O_2}^F) \mathbf{N} \Phi_{O_2}^e (1 - \mathbf{N} \Phi_{SiO_2}^e)) \end{array} \right) d\Omega^e \quad (A2)$$

$$\mathbf{F}_u^{e(F)} = \int_{\Omega^e} \mathbf{B}^T (\mathbf{N} \Phi_{SiO_2}^e \mathbf{C}_{O_2}^{SiO_2} + (1 - \mathbf{N} \Phi_{SiO_2}^e) \mathbf{C}_O^F) \mathbf{B} u^e d\Omega^e \quad (A3)$$

$$\mathbf{K}_{\phi_{O_2} \phi_{O_2}}^{e(F)} = - \int_{\Omega^e} \left(\begin{array}{c} \mathbf{N}^T \mathbf{N} \Delta s^F \mathbf{N} \frac{d\Phi_{SiO_2}^e}{dt} \\ + \mathbf{N}^T \mathbf{N} s_{O_2} \frac{1}{\Delta t} \\ + \nabla \mathbf{N}^T \nabla \mathbf{N} s_{O_2} D_{O_2} \\ + 3 \mathbf{N}^T \mathbf{N} (R_c s_{SiO_2} s_{O_2} (1 - \mathbf{N} \Phi_{SiO_2}^e)) \end{array} \right) d\Omega^e \quad (A4)$$

$$\mathbf{K}_{\phi_{O_2} \phi_{SiO_2}}^{e(F)} = - \int_{\Omega^e} \begin{pmatrix} \mathbf{N}^T \mathbf{N} \phi_{O_2}^e \Delta s^F \mathbf{N} \frac{1}{\Delta t} \\ + \mathbf{N}^T \mathbf{N} \Delta s^F \mathbf{N} \frac{d\phi_{O_2}^e}{dt} \\ + \nabla \mathbf{N}^T \nabla \mathbf{N} \phi_{O_2}^e \left(3 \Delta s^F \left(\mathbf{N} \phi_{SiO_2}^e \right)^2 \mathbf{N} D^0 + D_{\min O_2}^F \Delta s^F \mathbf{N} + 2 s_{\min O_2}^F \left(\mathbf{N} \phi_{SiO_2}^e \right) \mathbf{N} D^0 \right) \\ + 3 \mathbf{N}^T \mathbf{R}_c s_{SiO_2} \left(- s_{\min O_2}^F \mathbf{N} \phi_{O_2}^e \mathbf{N} + \mathbf{N} \Delta s^F \mathbf{N} \phi_{O_2}^e - 2 \mathbf{N} \Delta s^F \mathbf{N} \phi_{SiO_2}^e \mathbf{N} \phi_{O_2}^e \right) \end{pmatrix} d\Omega^e \quad (A5)$$

$$\mathbf{K}_{\phi_{O_2} \mathbf{u}}^{e(F)} = \mathbf{0} \quad (A6)$$

$$\mathbf{K}_{\phi_{SiO_2} \phi_{O_2}}^{e(F)} = - \int_{\Omega^e} \begin{pmatrix} \mathbf{0} \\ -2 \mathbf{N}^T \mathbf{N} \mathbf{R}_c s_{SiO_2} s_{O_2} \left(1 - \mathbf{N} \phi_{SiO_2}^e \right) \end{pmatrix} d\Omega^e \quad (A7)$$

$$\mathbf{K}_{\phi_{SiO_2} \phi_{SiO_2}}^{e(F)} = - \int_{\Omega^e} \begin{pmatrix} \mathbf{N}^T \mathbf{N} s_{SiO_2} \frac{1}{\Delta t} \\ -2 \mathbf{N}^T \mathbf{N} \mathbf{R}_c s_{SiO_2} \left(\Delta s^F \mathbf{N} \phi_{O_2}^e - 2 \Delta s^F \mathbf{N} \phi_{O_2}^e \mathbf{N} \phi_{SiO_2}^e - s_{\min O_2}^F \mathbf{N} \phi_{O_2}^e \right) \end{pmatrix} d\Omega^e \quad (A8)$$

$$\mathbf{K}_{\phi_{SiO_2} \mathbf{u}}^{e(F)} = \mathbf{K}_{\mathbf{u} \phi_{O_2}}^{e(F)} = \mathbf{0} \quad (A9)$$

$$\mathbf{K}_{\mathbf{u} \phi_{SiO_2}}^{e(F)} = - \int_{\Omega^e} \mathbf{B}^T \left(\mathbf{C}_o^{SiO_2} - \mathbf{C}_o^F \right) \mathbf{B} \mathbf{u}^e \mathbf{N} d\Omega^e \quad (A10)$$

$$\mathbf{K}_{\mathbf{u} \mathbf{u}}^{e(F)} = - \int_{\Omega^e} \mathbf{B}^T \left(\mathbf{N} \phi_{SiO_2}^e \mathbf{C}_o^{SiO_2} + \left(1 - \mathbf{N} \phi_{SiO_2}^e \right) \mathbf{C}_o^F \right) \mathbf{B} d\Omega^e \quad (A11)$$

The equations of the SiC matrix are as follows.

$$\mathbf{F}_{\phi_{O_2}}^{e(M)} = \int_{\Omega^e} \begin{pmatrix} \mathbf{N}^T \mathbf{N} \phi_{O_2}^e \left(\frac{\partial d}{\partial \epsilon} \mathbf{B} \frac{d\mathbf{u}^e}{dt} \right) \Delta s^M \\ + \mathbf{N}^T \mathbf{N} \left(\Delta s^M d + s_{\min O_2}^M \right) \frac{d\phi_{O_2}^e}{dt} \\ + \nabla \mathbf{N}^T \nabla \mathbf{N} \phi_{O_2}^e \left(\Delta D^M d + D_{\min O_2}^M \right) \left(\Delta s^M d + s_{\min O_2}^M \right) \\ + \mathbf{0} \end{pmatrix} d\Omega^e \quad (A12)$$

$$\mathbf{F}_{\phi_{SiO_2}}^{e(M)} = \mathbf{0} \quad (A13)$$

$$\mathbf{F}_{\mathbf{u}}^{e(M)} = \int_{\Omega^e} \mathbf{B}^T (1 - d) \mathbf{C}_o^M \mathbf{B} \mathbf{u}^e d\Omega^e \quad (A14)$$

$$\mathbf{K}_{\phi_{O_2} \phi_{O_2}}^{e(M)} = - \int_{\Omega^e} \begin{pmatrix} \mathbf{N}^T \mathbf{N} \left(\frac{\partial d}{\partial \epsilon} \mathbf{B} \frac{d\mathbf{u}^e}{dt} \right) \Delta s^M \\ + \mathbf{N}^T \mathbf{N} s_{O_2} \frac{1}{\Delta t} \\ + \nabla \mathbf{N}^T \nabla \mathbf{N} D_{O_2} s_{O_2} \\ + \mathbf{0} \end{pmatrix} d\Omega^e \quad (A15)$$

$$\mathbf{K}_{\phi_{O_2} \phi_{SiO_2}}^{e(M)} = \mathbf{0} \quad (A16)$$

$$\mathbf{K}_{\phi_{O_2} \mathbf{u}}^{e(M)} = - \int_{\Omega^e} \begin{pmatrix} \mathbf{N}^T \mathbf{N} \phi_{O_2}^e \left(\frac{\partial d}{\partial \epsilon} \mathbf{B} \frac{1}{\Delta t} \right) \Delta s^M \\ + \mathbf{N}^T \mathbf{N} \frac{d\phi_{O_2}^e}{dt} \left(\Delta s^M \frac{\partial d}{\partial \epsilon} \mathbf{B} \right) \\ + \nabla \mathbf{N}^T \nabla \mathbf{N} \phi_{O_2}^e \left(D_{O_2} \Delta s^M + s_{O_2} \Delta D^M \right) \frac{\partial d}{\partial \epsilon} \mathbf{B} \\ + \mathbf{0} \end{pmatrix} d\Omega^e \quad (A17)$$

$$\mathbf{K}_{\phi_{SiO_2} \phi_{O_2}}^{e(M)} = \mathbf{K}_{\phi_{SiO_2} \phi_{SiO_2}}^{e(M)} = \mathbf{K}_{\phi_{SiO_2} \mathbf{u}}^{e(M)} = \mathbf{0} \quad (A18)$$

$$\mathbf{K}_{\mathbf{u} \phi_{O_2}}^{e(M)} = \mathbf{K}_{\mathbf{u} \phi_{SiO_2}}^{e(M)} = \mathbf{0} \quad (A19)$$

$$\mathbf{K}_{uu}^{e(M)} = - \int_{\Omega^e} \mathbf{B}^T (1-d) \mathbf{C}_o^M \mathbf{B} - \mathbf{B}^T \mathbf{C}_o^M \mathbf{B} u^e \frac{\partial d}{\partial \epsilon} \mathbf{B} d\Omega^e \quad (\text{A20})$$

The equations of the porosity are as follows.

$$\mathbf{F}_{\phi_{O_2}}^{e(\text{Porosity})} = \int_{\Omega^e} \begin{pmatrix} \mathbf{0} \\ +\mathbf{N}^T \mathbf{N} s_{O_2} \frac{d\phi_{O_2}^e}{dt} \\ +\nabla \mathbf{N}^T \nabla \mathbf{N} D_{O_2} s_{O_2} \phi_{O_2}^e \\ +\mathbf{0} \end{pmatrix} d\Omega^e \quad (\text{A21})$$

$$\mathbf{F}_{\phi_{SiO_2}}^{e(\text{Porosity})} = \mathbf{0} \quad (\text{A22})$$

$$\mathbf{F}_u^{e(\text{Porosity})} = \int_{\Omega^e} \mathbf{B}^T \mathbf{C}_o^{\text{Porosity}} \mathbf{B} u^e d\Omega^e \quad (\text{A23})$$

$$\mathbf{K}_{\phi_{O_2} \phi_{O_2}}^{e(\text{Porosity})} = - \int_{\Omega^e} \begin{pmatrix} \mathbf{0} \\ +\mathbf{N}^T \mathbf{N} s_{O_2} \frac{1}{\Delta t} \\ +\nabla \mathbf{N}^T \nabla \mathbf{N} D_{O_2} s_{O_2} \\ +\mathbf{0} \end{pmatrix} d\Omega^e \quad (\text{A24})$$

$$\mathbf{K}_{\phi_{O_2} \phi_{SiO_2}}^{e(\text{Porosity})} = \mathbf{K}_{\phi_{O_2} u}^{e(\text{Porosity})} = \mathbf{0} \quad (\text{A25})$$

$$\mathbf{K}_{\phi_{SiO_2} \phi_{O_2}}^{e(\text{Porosity})} = \mathbf{K}_{\phi_{SiO_2} \phi_{SiO_2}}^{e(\text{Porosity})} = \mathbf{K}_{\phi_{SiO_2} u}^{e(\text{Porosity})} = \mathbf{0} \quad (\text{A26})$$

$$\mathbf{K}_{u \phi_{O_2}}^{e(\text{Porosity})} = \mathbf{K}_{u \phi_{SiO_2}}^{e(\text{Porosity})} = \mathbf{0} \quad (\text{A27})$$

$$\mathbf{K}_{uu}^{e(\text{Porosity})} = - \int_{\Omega^e} \mathbf{B}^T \mathbf{C}_o^{\text{Porosity}} \mathbf{B} d\Omega^e \quad (\text{A28})$$

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jeurceramsoc.2025.117335](https://doi.org/10.1016/j.jeurceramsoc.2025.117335).

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